

NAB2 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NAB2 gene fusions from the msk_impact_50k_2026 study, covering 72 fusion events. 25/72 fusions (34.7%) were in-frame. 49/72 fusions (68.1%) retained the PF04905 domain. Domain retention was tested with Fisher's exact test ($p=0.11258$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for NAB2.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 72 analyzed fusion events, identifying 14 distinct fusion partner genes. The most frequent partner was STAT6, observed in 53 events.

1.2 Domain retention

PF04905 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 20/25 (80.0%) of in-frame fusions retained the domain, $p=0.11258$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

1.3 Domain disruption

Domain-disruption analysis was skipped: NAB2 has no disruption_required_domains configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 71 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 372 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is 6 aa from the nearest configured domain boundary at 366 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: NAB2 has no group_field configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

1.8 Joint partner

Joint partner reported: NAB2 has no gene_pair configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
STAT6	53	0.7361111111111112	0.20043213737826426		0.3434207302131831		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.460649253156967	1
CUX2	1	0.013888888888888888	0.20043213737826426		0.948051948051948		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2806892760271785	2
PDZRN4	1	0.013888888888888888	0.20043213737826426		0.948051948051948		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2806892760271785	3
MGAT4C	2	0.02777777777777776	0.20043213737826426		0.9090909090909091		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27829286291647964	4
KRT6A	1	0.013888888888888888	0.20043213737826426		0.935064935064935		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2779063446728186	5
LRP1	3	0.04166666666666664	0.20043213737826426		0.8658008658008658		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2749688060209942	6
TMEM117	2	0.02777777777777776	0.20043213737826426		0.6558441558441559		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22402570150646114	7
KIF5A	1	0.013888888888888888	0.20043213737826426		0.6623376623376623		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.21946478623126012	8
FMN1	3	0.04166666666666664	0.20043213737826426		0.23376623376623373		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.13953281344214447	9
DLGAP2	1	0.013888888888888888	0.20043213737826426		0.23376623376623373		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12762805153738258	10
FAM186A	1	0.013888888888888888	0.20043213737826426		0.23376623376623373		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12762805153738258	11
LFNG	1	0.013888888888888888	0.20043213737826426				['domain_retention', 'recurrence']	0.09868127456587769	12
LGR5	1	0.013888888888888888	0.20043213737826426		0.09740259740259738		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.09840727231660334	13
MTOR	1	0.013888888888888888	0.20043213737826426		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07753528715890391	14

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
319	0	49	3	19	0.0	0.02694427434170239	1.5695335082504807
320	0	49	4	18	0.0	0.00752854724253449	2.123288820003888
352	0	49	5	17	0.0	0.0020225947815764307	2.6940911176014084
364	0	49	13	9	0.0	8.560748975173139e-09	8.067488237477606
372	0	49	14	8	0.0	1.3283920823544527e-09	8.876673721601218
381	1	48	14	8	0.011904761904761904	1.731570731911155e-08	7.761559763513135
382	3	46	15	7	0.030434782608695653	1.0398205008988075e-07	6.983041624352472
400	4	45	15	7	0.04148148148148148	4.330878206429079e-07	6.36342402938942
426	4	45	16	6	0.03333333333333333	7.233726445978198e-08	7.140637918749475
434	5	44	16	6	0.04261363636363636	2.7039639334463545e-07	6.567999105485465
490	45	4	20	2	1.125	1.0	-0.0
502	46	3	20	2	1.5333333333333334	0.6419888956213136	0.19247248378799386
508	47	2	20	2	2.35	0.5828392348978783	0.23445122058408072
510	48	1	20	2	4.8	0.2249846907532149	0.6478470327668787
511	48	1	21	1	2.2857142857142856	0.5267605633802818	0.2783867465185951
521	49	0	21	1		0.3098591549295775	0.508835667896869

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20	29
5	17

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
auto_key_domain	NAB conserved region 2 (NCD2)	71	49	13	9	0.35547087642896025

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
CUX2	1
DLGAP2	1
FAM186A	1
FMN1	3
KIF5A	1
KRT6A	1
LFNG	1
LGR5	1
LRP1	3
MGAT4C	2
MTOR	1
PDZRN4	1
STAT6	53
TMEM117	2

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_pos	breakpoint_pos	breakpoint_pos	protein_pos	protein_pos	breakpoint_pos	status	main_ref	domain	domain	disrupted_domains	domains_annotation	annotation_effect_on_fusion
EVT-P-0052994-T01-IM6-1	P-0052994-T01-IM6		FAM186A-NAB2	FAM186A	out-of-frame	three_prime	57488361	7	490	True	lost	0.0	False		NAB conserved region 2 (NCD2)			FAM186A (NM_001145475) - NAB2 (NM_005967) rearrangement: c.193-15726:FAM186A_c.1469-34:NAB2inv Protein Fusion: out of frame {FAM186A;NAB2}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	breakpoint_5	main_ref	donor_frame	is_maintained	domain	disrupted_domain	description_annotation	status	effect_on_fra
EVT-P-0004282-T03-IM5-62	P-0004282-T03-IM5		NAB2-STAT6	STAT6	in-frame	five_prime	57488115	7	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)				NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exon 16 to 22): c.1469-280:NAB2_c.1745-157:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA (NM_005967)	
EVT-P-0033539-T02-IM6-63	P-0033539-T02-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57486902	5	400	False	retained	1.0	False	NAB conserved region 2 (NCD2)				NAB2 (NM_005967) rearrangement: c.1200:NAB2_chr12:g.57486902inv Protein Fusion: mid-exon {NAB2:STAT6}	NA (NM_005967)	
EVT-P-0019357-T03-IM6-64	P-0019357-T03-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57488027	7	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)				NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1469-368:NAB2_c.1690:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA (NM_005967)	
EVT-P-0019182-T01-IM6-66	P-0019182-T01-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57487665	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)				NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused in frame with STAT6 exons 16-22): c.1468+284:NAB2_c.1744+39:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA (NM_005967)	
EVT-P-0036288-T01-IM6-67	P-0036288-T01-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57488128	7	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)				NAB2 (NM_005967) - STAT6 (NM_003153) fusion: c.1469-267:NAB2_c.1850:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA (NM_005967)	
EVT-P-0036975-T01-IM6-68	P-0036975-T01-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57486781	4	381	True	retained	1.0	False	NAB conserved region 2 (NCD2)				NAB2 (NM_005967) rearrangement: c.1143+30:NAB2_chr12:g.57508083inv Transcript Fusion {NAB2:STAT6}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0004405-T01-IM5-74	P-0004405-T01-IM5		NAB2-STAT6	STAT6	unknown	five_prime	57487215	6	434	False	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - Fusion (NAB2 exon 6 fused to STAT6 exon 18) : c.1302:NAB2_c.1969:STAT6inv	NA (NM_003153)	
EVT-P-0004943-T01-IM5-75	P-0004943-T01-IM5		NAB2-STAT6	STAT6	unknown	five_prime	57488487	7	521	False	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) reciprocal fusion (NAB2 exons 1-7 with STAT6 exons 15-22) : c.1561:NAB2_c.1739:STAT6inv	NA (NM_005967) - STAT6 (NM_003153)	
EVT-P-0040364-T01-IM6-76	P-0040364-T01-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57487738	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+357:STAT6inv	NA (NM_005967) - STAT6 (NM_003153)	
EVT-P-0040364-T02-IM6-77	P-0040364-T02-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57487738	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			Fusion: in frame {NAB2:STAT6} NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+357:STAT6inv	NA (NM_005967) - STAT6 (NM_003153)	
EVT-P-0014386-T01-IM6-78	P-0014386-T01-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57487858	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			Fusion: in frame {NAB2:STAT6} NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 17 - 22): c.1468+477:STAT6inv	NA (NM_005967) - STAT6 (NM_003153)	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_stat	target_role	breakpoint	breakpoint	breakpoint	protein_pos	domain_pos	main_retained	donor_inframe	donor_deleted	host_domain	disrupted_domains	annotation	site_effect_on_fra
EVT-P-0043579-T01-IM6-79	P-0043579-T01-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57488047	7	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-348:NAB2_c.1744+96:STAT6inv Protein Fusion: in frame {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-358:NAB2_c.1744+93:STAT6inv Protein Fusion: in frame {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-126:NAB2_c.1700:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+2:NAB2_c.1769:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+399:NAB2_c.1789:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92:NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	
EVT-P-0044036-T01-IM6-80	P-0044036-T01-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57488037	7	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-358:NAB2_c.1744+93:STAT6inv Protein Fusion: in frame {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-358:NAB2_c.1744+93:STAT6inv Protein Fusion: in frame {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-126:NAB2_c.1700:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+2:NAB2_c.1769:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+399:NAB2_c.1789:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92:NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	
EVT-P-0044232-T02-IM6-81	P-0044232-T02-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57488269	7	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-126:NAB2_c.1700:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+2:NAB2_c.1769:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+399:NAB2_c.1789:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92:NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	
EVT-P-0001947-T03-IM6-82	P-0001947-T03-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487383	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+2:NAB2_c.1769:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+399:NAB2_c.1789:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92:NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	
EVT-P-0045590-T01-IM6-83	P-0045590-T01-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487780	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+399:NAB2_c.1789:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92:NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	
EVT-P-0046398-T01-IM6-84	P-0046398-T01-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487473	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92:NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6} NAB2 NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92:NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	
EVT-P-0046398-T02-IM6-85	P-0046398-T02-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487473	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92:NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	is_main_ref	is_in_frame	is_in_domain	is_disrupted	description	annotation	effect_on_fra
EVT-P-0006243-T01-IM5-103	P-0006243-T01-IM5		NAB2-STAT6	STAT6	unknown	five_prime	57488431	7	502	False	retained	1.0	False	NAB conserved region 2 (NCD2)		NAB2(NM_005967) - STAT6(NM_003153) Fusion (NAB2 exons 1 to 7 fused with STAT6 exon 16 to 22) : c.1505:NAB2_c.1745-142:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA (NM_005967) - STAT6 (NM_003153)	
EVT-P-0005578-T01-IM5-104	P-0005578-T01-IM5		NAB2-STAT6	STAT6	in-frame	five_prime	57486815	5	382	True	retained	1.0	False	NAB conserved region 2 (NCD2)		NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exon4 fused with STAT6 exon5) : c.1144-31:NAB2_c.340-81:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+192:NAB2_c.1861:STAT6inv	
EVT-P-0067841-T01-IM7-106	P-0067841-T01-IM7		NAB2-STAT6	STAT6	unknown	five_prime	57487573	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)		NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+192:NAB2_c.1861:STAT6inv	NA (NM_005967) - STAT6 (NM_003153) fusion: mid-exon {NAB2:STAT6}	
EVT-P-0066305-T01-IM7-107	P-0066305-T01-IM7		NAB2-STAT6	STAT6	in-frame	five_prime	57488138	7	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)		NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-257:NAB2_c.1745-14:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1468+71:NAB2_c.1744+14:STAT6inv	
EVT-P-0027038-T01-IM6-108	P-0027038-T01-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57487452	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)		NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1468+71:NAB2_c.1744+14:STAT6inv	NA (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1468+71:NAB2_c.1744+14:STAT6inv	

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3 Figures

NAB2 domain-retention track

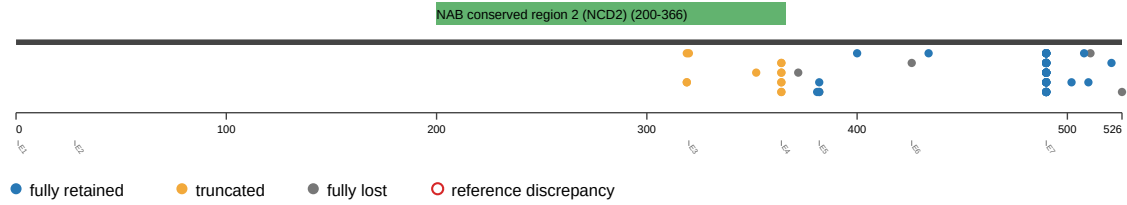


Figure 1: domain retention outliers

NAB2 fusion-transcript schematic

partner block → breakpoint → retained NAB2 portion (domain-colored, exon ticks)

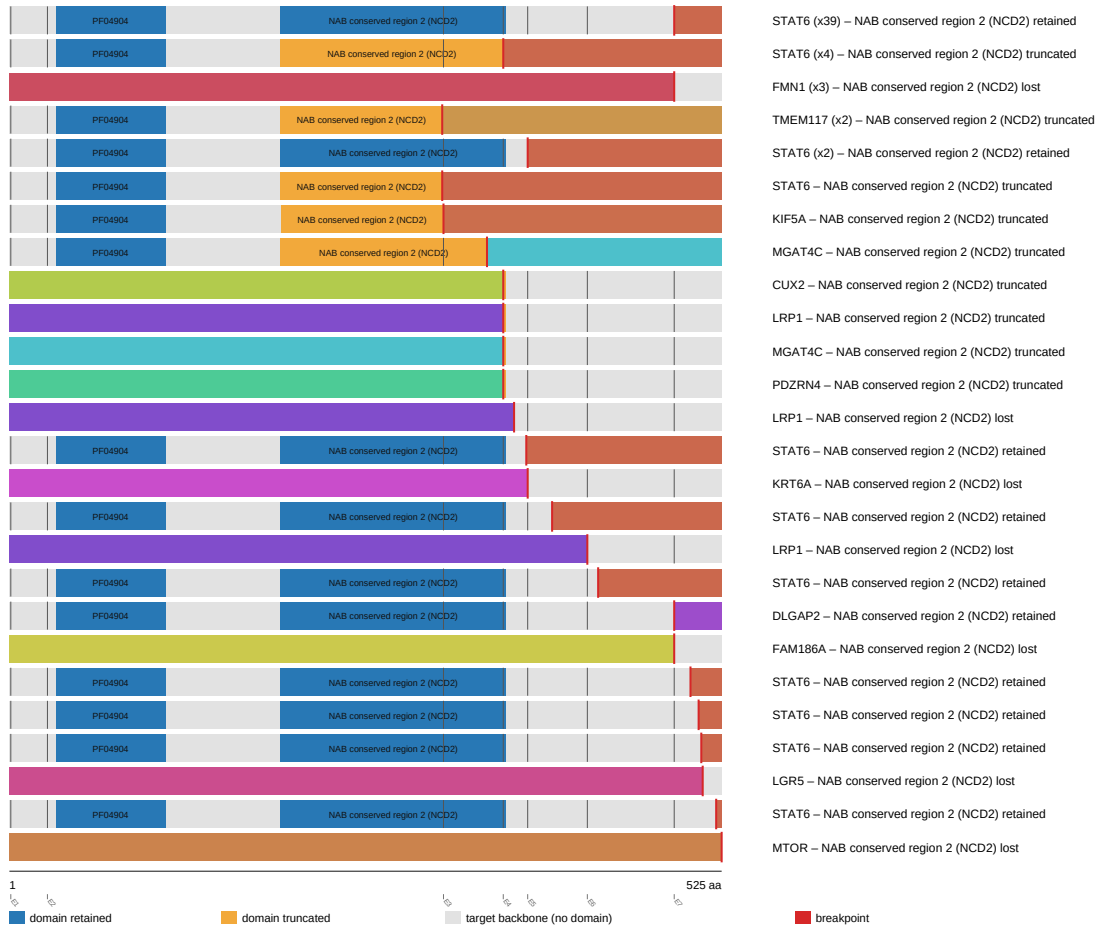


Figure 2: fusion schematic

NAB2 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison